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PADM: Parkinson's Analysis Through Digital Motor Metrics

“Success isn’t something that happens by chance; it’s built through the opportunity one creates for oneself.” is something my grandfather told me throughout my life. I’m proposing this project not only as an analysis of data, but as a homage to past family members. My goal with PADM (Parkinson’s Analysis Through Digital Motor Metrics) is to provide a program focused on detecting early signs of Parkinson’s Disease through subtle keyboard and touchpad interactions. This research will allow me to apply machine learning tools to practical problems and conduct in-depth data analysis to draw meaningful results. The tools I plan to use will primarily be drawn from my coursework in Intro to Artificial Intelligence and Information Retrieval, where I developed practical experience with machine learning and advanced data analysis methods. As for how this project affects my future, it will help prepare me for classes in my upcoming senior year, and provide me with real-world applications I can carry into my professional career. Annually, 60,000 people in the United States are diagnosed with Parkinson’s disease, and approximately 1,000,000 individuals are currently living with it. With the average age of diagnosis being 60 years old, my personal mission is to contribute to earlier detection methods that can improve quality of life, expand treatment opportunities, and ultimately provide individuals and families with more time and clarity in confronting the disease.

Referenced in the introduction above, PADM’s goal is to provide non-invasive early detection methodologies by analyzing fine motor skills and cognitive functionality. This project

will combine both research and creative development, as I build and refine the program's backend until I am confident enough in the data and results to expand it into a full application over the summer.

The program will begin by prompting the user for demographic and background information, including age, sex, handedness, overall sleep quality, and genetic predisposition to Parkinson's disease. It will then guide the user through a series of motor skill assessments, measuring tapping speed in each hand, dominant and non-dominant tremor rates (in Hz), and tracing precision tasks performed through a keyboard and touchpad interface.

Following the motor assessments, the program will administer a set of cognitive aptitude tests designed to evaluate memory retention and processing speed. The collected data will then be analyzed and compared against reference data obtained from the Michael J. Fox Foundation, ultimately generating a similarity score based on observed patterns. The primary purpose of this program is not to serve as a diagnostic tool, but to provide comparative, data-driven insight aligned with established research patterns.

As the semester progresses, I plan to divide the project into four distinct phases running through May 13, 2026. Phase 1 (ongoing through March 6) consists of initial research, including reviewing relevant datasets, refining measurable variables, and structuring the backend data framework. Upon approval of this proposal, I will move directly into Phase 2 (March 7 – March 27), which will focus on developing the program's interface and implementing the motor skills assessments, including tapping speed, tremor rate measurements, and tracing precision tasks. Phase 3 (March 28 – April 17) will involve integrating the cognitive testing series, including memory and processing speed assessments, and refining the similarity scoring system. Phase 4

(April 18 – May 9) will serve as a revision and testing period, allowing me to refine inconsistencies, strengthen the scoring model, and improve overall functionality before final submission preparation (May 10 – May 13). Following the completion of Phase 4, I intend to expand this project into a fully developed application over the summer, building on the backend framework established during the semester.

Through this project, I aim to deepen my understanding of data-driven markers associated with early-stage Parkinson's disease while strengthening my ability to translate machine learning concepts into functional software. Specifically, I intend to develop skills in backend system design, feature extraction from motor input data, quantitative tremor and processing speed analysis, and similarity-based scoring methods. This project will require me to integrate motor and cognitive data into one analytical framework, refine algorithms through continuous testing, and document my methodology in a clear and insightful way. By the end of the semester, I expect to demonstrate technical proficiency in building a working prototype, analytical competence in interpreting its outputs, and a stronger foundation for advanced coursework and future research in applied machine learning.

This independent study represents both a technical challenge and a meaningful opportunity to apply artificial intelligence to a real-world health concern. By developing and refining PADM, I aim to contribute to accessible, data-driven early detection methodologies while strengthening my own analytical and programming skills. The structured phases, measurable deliverables, and final research analysis ensure that this project remains rigorous and accountable. Ultimately, this work lays the foundation for continued development and expansion beyond the semester.

Bibliography

Parkinson's Progression Markers Initiative. *PPMI*. Michael J. Fox Foundation,

<https://www.ppmi-info.org/>.

Accelerating Medicines Partnership Parkinson's Disease (AMP-PD). *AMP PD*.

<https://amp-pdrd.org/>.

The Michael J. Fox Foundation. *Fox Den*. <https://www.michaeljfox.org/fox-den>.

Parker, "PD BiostampRC21 Parkinson's Disease Accelerometry Dataset: Five Wearable Sensor Study." *IEEE Dataport*, IEEE,

<https://ieee-dataport.org/open-access/pd-biostamprc21-parkinsons-disease-accelerometry-dataset-five-wearable-sensor-study-0>.

The Michael J. Fox Foundation. "Introducing PARADIGM: An Open-Source Toolbox for Deriving PD Digital Biomarkers from Real-Life Wrist Sensor Data." *RCOP*,

<https://rcop.michaeljfox.org/t/introducing-paradigm-an-open-source-toolbox-for-deriving-pd-digital-biomarkers-from-real-life-wrist-sensor-data/1300>.

Sejdić, *PLOS Digital Health*, vol. X,

<https://journals.plos.org/digitalhealth/article?id=10.1371/journal.pdig.0000519>.

AI and Ethics, vol. X, <https://link.springer.com/article/10.1007/s00146-023-01843-6>.